UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. A linear presentation allows users to choose their own path.
2. Multimedia combines text, graphics, audio, video and animation.
3. A plotter is an input device.
4. A web browser displays hypertext documents.
5. Chroma key compositing replaces a colour background with another image.
6. Hypermedia combines multimedia with hyperlinks.
7. HTML is used to structure content on web pages.
8. A scanner is an input device.
9. MIDI files store digitized audio samples.
10. A multimedia project always uses every media element.
11. RAM is a type of permanent storage.
12. A hyperlink can link only to another web page, not to a file.
13. A licence states how a work may be used.
14. Interactivity level is irrelevant when choosing an authoring tool.
15. Card-based authoring tools are best for messages with a clear beginning and end.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Browser	B. Target audience	C. Keyframe
D. Kerning	E. Card-based tool	F. Sampling
G. Bitmap	H. Storyboard	I. Amplitude
J. Frame rate	K. Serif	L. Sans serif
M. Tweening	N. Vector	O. MIDI

1. important frame in an animation
2. image made of pixels
3. converting sound into digital values
4. font with small strokes at character ends
5. standard for musical notes and instructions
6. font without small strokes
7. image made of mathematical objects
8. software that displays web pages
9. visual sequence plan
10. adjustment of space between character pairs
11. frames displayed per second
12. height of a sound wave
13. authoring tool organised like pages or cards
14. intended users of a product
15. generating in-between frames

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PART C (30 MARKS)

Choose the correct answer for each question.

1. MPEG is:
   A. A font
   B. An image editor
   C. A video compression family
   D. A storage device

2. Which is an authoring tool category?
   A. Card-based
   B. JPEG
   C. RGB
   D. CMYK

3. Which is a 3D animation task?
   A. Sampling
   B. Modelling and rendering
   C. Kerning
   D. Bookmarking

4. Hertz measures:
   A. Loudness
   B. File size
   C. Resolution
   D. Frequency

5. Morphing changes:
   A. Font size
   B. One shape into another
   C. File format
   D. Audio pitch

6. RAM is:
   A. Permanent storage
   B. An input device
   C. Temporary working memory
   D. A web browser

7. Virtual reality creates:
   A. A printed page
   B. A text font
   C. A simulated environment
   D. A camera lens

8. Which format is commonly used for bitmap images?
   A. MPEG
   B. PNG
   C. HTML
   D. MIDI

9. Which is an output device?
   A. Scanner
   B. Mouse
   C. Plotter
   D. Keyboard

10. RGB is used mainly for:
    A. Video editing only
    B. Printed output
    C. Screen display
    D. Audio recording

11. A prototype is:
    A. An early model of a product
    B. A font
    C. A final release
    D. A codec

12. A bitmap image is made of:
    A. Sound samples
    B. Pixels
    C. Mathematical objects
    D. Text characters

13. Cel animation is created by:
    A. Drawing each frame by hand
    B. Scanning text
    C. Recording sound
    D. Printing images

14. An authoring tool is used to:
    A. Only browse the web
    B. Assemble multimedia elements
    C. Only record audio
    D. Only print text

15. Tweening creates:
    A. Text fonts
    B. Sound effects
    C. Video codecs
    D. In-between frames

16. Which best defines multimedia?
    A. A printed book
    B. Text only
    C. Text, graphics, audio, video and animation
    D. A single image

17. Streaming allows media to:
    A. Require full download first
    B. Only play offline
    C. Play while downloading
    D. Remove compression

18. Path animation moves an object:
    A. Without motion
    B. Randomly
    C. Only vertically
    D. Along a predefined route

19. Frame rate is:
    A. Frames per second
    B. Pixels per inch
    C. Samples per second
    D. Bits per second

20. Convergence in multimedia refers to:
    A. Deleting media files
    B. Recording audio only
    C. Printing images
    D. Merging of media and computing

21. Which is an input device?
    A. Scanner
    B. Printer
    C. Plotter
    D. Monitor

22. Interactivity in multimedia means:
    A. There is no navigation
    B. The system plays automatically
    C. The user can control content and flow
    D. Only text is shown

23. Which is a benefit of multimedia in education?
    A. It replaces all teachers
    B. It combines visuals and sound
    C. It removes all interaction
    D. It uses only text

24. Which colour model is subtractive?
    A. HSL
    B. RGB
    C. CMYK
    D. HSV

25. Anti-aliasing is used to:
    A. Add sound
    B. Increase file size only
    C. Compress video
    D. Smooth jagged edges

26. Hypermedia is best described as:
    A. Multimedia with hyperlinks
    B. Text-only media
    C. A printed manual
    D. A single video file

27. Interlacing:
    A. Draws alternate lines of a frame
    B. Adds sound
    C. Removes colour
    D. Draws every line at once

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28. A higher sampling rate generally:

A. Changes colour
B. Reduces accuracy
C. Improves accuracy
D. Removes audio

29. Which is an example of linear multimedia?

A. An interactive kiosk
B. A cinema presentation
C. A game with choices
D. A website with menus

30. Which image type scales without losing quality?

A. Scan
B. Bitmap
C. Photograph
D. Vector

PART D (40 MARKS)

QUESTION 1

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(3 marks)

b) Describe TWO roles of hyperlinks in the website.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(4 marks)

QUESTION 2

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

(2 marks)

b) Describe TWO roles of hypertext in multimedia.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

(5 marks)

QUESTION 3

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

(4 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

QUESTION 4

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(3 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(3 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

END OF QUESTION PAPER